

cDFT_electrolytes

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1 Introduction

The Helmholtz free energy functional $\mathcal{F}$ is given by:

$$\mathcal{F}[\{\rho_i\}, \vec{P}] = \mathcal{F}_{\text{ideal}}[\{\rho_i\}, \vec{P}] + \mathcal{F}_{\text{excess}}[\{\rho_i\}, \vec{P}] + \mathcal{F}_{\text{external}}[\{\rho_i\}, \vec{P}] \quad (1)$$

We can further decompose this functional into two parts:

$$\mathcal{F}[\{\rho_i\}, \vec{P}] = \mathcal{F}^{(0)}[\{\rho_i\}] + \mathcal{F}^{(1)}[\{\rho_i\}, \vec{P}], \text{ where} \quad (2)$$

$$\mathcal{F}^{(0)}[\{\rho_i\}] = \mathcal{F}_{\text{ideal}}^{(0)}[\{\rho_i\}] + \mathcal{F}_{\text{excess}}^{(0)}[\{\rho_i\}] + \mathcal{F}_{\text{external}}^{(0)}[\{\rho_i\}] \text{ and} \quad (3)$$

$$\mathcal{F}^{(1)}[\{\rho_i\}, \vec{P}] = \mathcal{F}_{\text{ideal}}^{(1)}[\rho_0, \vec{P}] + \mathcal{F}_{\text{excess}}^{(1)}[\{\rho_i\}, \vec{P}] + \mathcal{F}_{\text{external}}^{(1)}[\{\rho_i\}, \vec{P}] \quad (4)$$

Some of these contributions are standard:

$$\mathcal{F}_{\text{ideal}}^{(0)}[\{\rho_i\}] = \beta^{-1} \sum_i \int \rho_i(\vec{x}) \left(\ln \rho_i(\vec{x}) - 1 \right) d\vec{x} \quad (5)$$

$$\mathcal{F}_{\text{external}}^{(0)}[\{\rho_i\}] = \sum_i \int \rho_i(\vec{x}) V_{\text{ext},i}(\vec{x}) d\vec{x} \quad (6)$$

$$\mathcal{F}_{\text{ideal}}^{(1)}[\{\rho_i\}, \vec{P}] = \beta^{-1} \int \rho_0(\vec{x}) \left[\ln \left(\frac{\mathcal{L}^{-1}(\Omega(\vec{x}))}{\sinh(\mathcal{L}^{-1}(\Omega(\vec{x})))} \right) + \Omega(\vec{x}) \mathcal{L}^{-1}(\Omega(\vec{x})) \right] \quad (7)$$

Here, $\Omega(\vec{x}) = \frac{P(\vec{x})}{p\rho_0(\vec{x})}$ (where, $P(\vec{x}) = |\vec{P}(\vec{x})|$) and $\mathcal{L}(x) = \coth(x) - \frac{1}{x}$ is the Langevin function.

2 Update equation (potential balance)

2.1 The densities

For the i -th component, the update looks like:

$$\rho_i(\vec{x}) = \exp\left\{-\beta V_{\text{ext}}(\vec{x}) + \beta \mu_i + c_i^{(1)}(\vec{x})\right\} \quad (8)$$

where the one-body effective potential $c_i^{(1)}(\vec{x})$ are given by:

$$c_i^{(1)}(\vec{x}) = -\beta \frac{\delta}{\delta \rho_i} \left[\mathcal{F}_{\text{excess}}^{(0)}[\{\rho_i\}] + \mathcal{F}_{\text{excess}}^{(1)}[\{\rho_i\}, \vec{P}] + \mathcal{F}_{\text{ideal}}^{(1)}[\rho_0, \vec{P}] \right] \quad (9)$$

Now, we have

$$\mathcal{F}_{\text{excess}}^{(1)}[\{\rho_i\}, \vec{P}] = \frac{1}{2} \int \rho_q(\vec{x}) \phi(\vec{x}) d\vec{x} \quad (10)$$

where $\rho_q(\vec{x}) = \sum_i Q_i \rho_i(\vec{x}) - \nabla \cdot \vec{P}(\vec{x})$ is the source charge distribution and $\phi(\vec{x})$ is the electrostatic potential generated by it using the Poisson equation:

$$\nabla^2 \phi(\vec{x}) = -\frac{\rho_q(\vec{x})}{\epsilon_0} \quad (11)$$

If we include the boundary conditions while computing the electrostatic potential and the corresponding electric field ($\vec{E}(\vec{x}) = -\nabla \phi(\vec{x})$), we can safely choose $\mathcal{F}_{\text{external}}^{(1)}[\{\rho_i\}, \vec{P}] = 0$.

This leads to:

$$\frac{\delta}{\delta \rho_i} \mathcal{F}_{\text{excess}}^{(1)}[\{\rho_i\}, \vec{P}] = Q_i \phi(\vec{x}) \quad (12)$$

and furthermore, $\mathcal{F}_{\text{excess}}^{(0)}[\{\rho_i\}]$ can be decomposed into contributions arising due to various terms:

$$\mathcal{F}_{\text{excess}}^{(0)}[\{\rho_i\}] = \mathcal{F}_{\text{excess,HS}}^{(0)}[\{\rho_i\}] + \mathcal{F}_{\text{excess,LJ}}^{(0)}[\{\rho_i\}] \text{ where,} \quad (13)$$

$$\mathcal{F}_{\text{excess,HS}}^{(0)}[\{\rho_i\}] = \int \Phi_{\text{HS}}(\{n_\alpha(\vec{x})\}) d\vec{x} \quad (14)$$

$$\mathcal{F}_{\text{excess,LJ}}^{(0)}[\{\rho_i\}] = \frac{1}{2} \sum_{i,j} \int \rho_i(\vec{x}_1) \rho_j(\vec{x}_2) U_{\text{LJ}}^{i,j}(|\vec{x}_1 - \vec{x}_2|) \quad (15)$$

This leads to:

$$\frac{\delta}{\delta \rho_i} \mathcal{F}_{\text{excess,HS}}^{(0)}[\{\rho_i\}] = \sum_\alpha \int \frac{\partial \Phi_{\text{HS}}}{\partial n_\alpha(\vec{x}_1)} \frac{\delta n_\alpha(\vec{x}_1)}{\delta \rho_i(\vec{x})} d\vec{x}_1 \quad (16)$$

$$\frac{\delta}{\delta \rho_i} \mathcal{F}_{\text{excess,LJ}}^{(0)}[\{\rho_i\}] = \sum_j \int \rho_j(\vec{x}_1) U_{\text{LJ}}^{i,j}(|\vec{x}_1 - \vec{x}|) d\vec{x}_1 \quad (17)$$

and finally, the contribution to the solvent density due to density-polarization coupling through the $\mathcal{F}_{\text{ideal}}^{(1)}[\rho_0, \vec{P}]$ term :

$$\frac{\delta}{\delta \rho_0} \mathcal{F}_{\text{ideal}}^{(1)}[\rho_0, \vec{P}] = \ln \left(\frac{\mathcal{L}^{-1}(\Omega(\vec{x}))}{\sinh(\mathcal{L}^{-1}(\Omega(\vec{x})))} \right) \quad (18)$$

2.2 The polarization

$$\vec{P}(\vec{x}) = p\rho_0(\vec{x})\mathcal{L}\left(\beta p|\vec{E}(\vec{x})|\right)\hat{E}(\vec{x}) \quad (19)$$

3 Translational invariance - 1D geometry

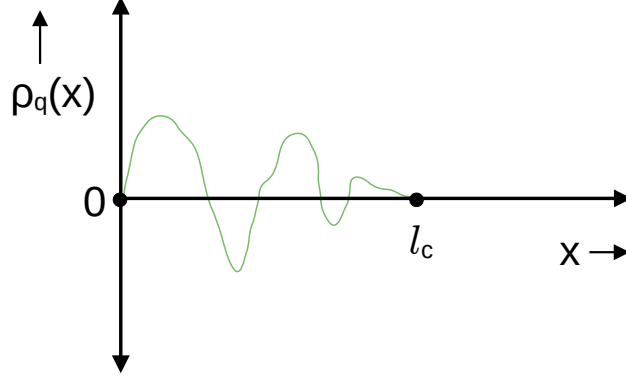


Figure 1: source-charge density profile in 1D

When the observables vary only along one of the directions (say x), i.e., the perpendicular direction to the wall, The electrostatic contributions simplify greatly. For example, if we are only interested in systems where the charge density $\rho_{q,\text{free}}(x) (= \sum_i Q_i \rho_i(x))$ and polarization $P(x)$ becomes zero beyond some $x = l_c$ from the wall (see above picture). Then for $x > l_c$, we have:

$$\phi''(x) = 0$$

$$\implies \phi(x) = Ax + B$$

If we set the gauge $\phi(\infty) = \phi_R$, we are constrained to have $A = 0$ (this means $E(\infty) = 0$), $B = \phi_R$. That implies the $\phi(x) = \phi_R$ and hence $E(x) = 0$ for $x > l_c$. For $x < l_c$, we have:

$$\phi(x) = \phi_R - \frac{1}{\epsilon_0} \int_x^\infty (y - x) \rho_{q,\text{free}}(y) dy - \frac{1}{\epsilon_0} \int_x^\infty P(y) dy \quad (20)$$

$$E(x) = -\phi'(x) = -\frac{1}{\epsilon_0} \int_x^\infty \rho_{q,\text{free}}(y) dy - \frac{1}{\epsilon_0} P(x) \quad (21)$$

We have gauge freedom so we choose $\phi_R = 0$. So it is fairly straightforward to use these formulas to generate the ES contributions from previous $\rho_i(x), P(x)$ profiles if our implementation box is large enough that $x = L$ can be treated as $x = \infty$ (or in other words $l_c < L$)